XVatsim Freeware User Guide

Version 2.2.0

Updated: October 2026

XVatsim is a Windows plugin for X-Plane 12 and xPilot. It gives VATSIM pilots a clean, route-aware frequency overlay that focuses on the controllers relevant to the current IFR flight.

XVatsim is intended for home flight simulation only. It is not approved for real-world aviation, navigation, flight planning, dispatch, or air traffic control use.



Clean XVatsim overlay

What's New In Version 2.2.0

Version 2.2.0 strengthens the Brain-owned controller decision system. Hard facts such as the live VATSIM roster, VATSpy mapping, valid radio service, transmitter distance, route sectors, and departure or arrival terminal polygons can now outweigh weaker callsign spelling differences. Similar callsigns such as the established Australian abbreviations remain useful positive evidence. Workers continue to report facts only; the Brain remains the only module that decides which controllers appear.

Departure and arrival terminal mapping now follows the applicable polygon stack for those two airports. It can recognize a specific inner Approach/Departure area, such as THAMES at Heathrow, while retaining the larger parent sector when the inner area is not active. This improves nested terminal coverage without processing unrelated terminal polygons around the world.

Standby Assist now uses regional arrival sequencing. Airports with CTAF use Center, Approach/Departure, Tower, CTAF, then Ground. Airports without CTAF use Center, Approach/Departure, UNICOM `122.800`, Tower, then Ground. During Enroute, Assist can place UNICOM `122.800` in COM1 standby when no eligible controller is online. Aircraft electrical-state handling and diagnostic log limits were also strengthened for more custom-aircraft compatibility and bounded disk use.

The new Manager can retrieve a future XVatsim payload only after validating its HTTPS location, exact size, SHA-256 hash, safe contents, version, and every managed file. Installation still requires the pilot to select Install, Update, or Repair and uses the existing backup and rollback transaction. Pilots using the Version 2.1.0 Manager must download the Version 2.2.0 package once from X-Plane.org or GitHub Releases and run the new Manager included in the package.

Version 2.2.0 retains the read-only xPilot 4 companion, proven xPilot 3 path, Brain-owned METAR, ATIS, PDC, controller selection, and radio decisions from Version 2.1.0.

Version 2.0.0 Feature Foundation

Version 2.0.0 adds three information ORBs below the main frequency card: METAR, VATSIM ATIS, and an initial PDC snapshot from xPilot. Each ORB opens its own information drawer, keeps its own history, and marks newly received content as read only after that content is visibly displayed.

Controller selection is also stronger. In the United States, XVatsim can use vNAS sector information as another source of evidence when deciding which Approach or Departure controller serves the flight. Terminal-controller transmitter distance is evaluated from the departure or arrival airport rather than from the moving aircraft. Center display continues to use the aircraft's 250-nautical-mile range gate. These facts support the same central decision system; they do not create a second controller-selection path.

Version 2.0.0 also prevents identical ATIS content from repeatedly returning to unread, improves exact route-polygon crossing detection, and keeps expensive route and authority preparation away from the X-Plane flight loop so normal settled operation remains calm.

What XVatsim Does

XVatsim is not a replacement for xPilot. xPilot remains the VATSIM client. XVatsim watches the active VATSIM flight plan, aircraft state, radio state, reachable controller list, and route context, then displays the frequencies that are relevant to the current flight.

The goal is to reduce cockpit clutter. Instead of showing every controller that might be nearby, XVatsim focuses on the current phase of flight:

- Departure: departure airport local services, departure approach or departure control, current center, and CTAF or UNICOM fallback.
- Enroute: route-relevant center controllers.
- Arrival: destination-relevant center, approach, tower, ground, ATIS, CTAF, or UNICOM when the flight is close enough to arrival.

XVatsim can sleep when there is nothing useful to show and wake when controller or arrival context becomes relevant.

Requirements

- Windows
- X-Plane 12
- xPilot 3 or xPilot 4
- VATSIM account and active xPilot connection
- IFR flight plan filed on VATSIM

XVatsim Version 2.2.0 does not include Mac support, Linux support, X-Plane 11 support, SimBrief import, Navigraph AIRAC import, public radio-chat display, or a dedicated VFR controller-evidence workflow.

Installation

1. Close X-Plane 12 and xPilot.
2. Download and extract the XVatsim zip from the X-Plane.org file page or the

official GitHub Releases page at `github.com/ezsimulations/XVatsim/releases`.

3. Open the extracted folder and run `XVatsimManager.exe`.
4. The Manager is distributed as unsigned freeware. If Windows SmartScreen

appears, select `More info`, confirm the file name is `XVatsimManager.exe`, and select `Run anyway`.

5. Review the X-Plane and xPilot folders found by the Manager. If a folder was

not found, choose it manually. The Manager will accept only a validated installation folder.

6. Select `Install`, `Update`, or `Repair`, whichever the Manager offers.
7. Wait for the successful completion message, then close the Manager.
8. Start X-Plane 12 and xPilot, then connect to VATSIM.

The Manager automatically installs the xPilot 4 companion when a validated xPilot 4 installation is present. It does not install a companion for xPilot 3. If the plugin is installed correctly, the `XVatsim` menu appears under X-Plane's `Plugins` menu.

For a manual X-Plane-only installation, copy the included `Resources` folder into the X-Plane 12 root folder and confirm this file exists:

```
X-Plane 12\Resources\plugins\XVatsim\win_x64\XVatsim.xpl
```

The manual fallback does not install the xPilot 4 companion; xPilot 4 pilots should use the Manager for complete Version 2.2.0 support.

First Flight Workflow

1. File an IFR flight plan on VATSIM.
2. Start X-Plane 12 and load the aircraft.
3. Start xPilot and connect to VATSIM using the same callsign as the flight plan.
4. Allow XVatsim a moment to load the flight plan and route context.
5. Fly normally. XVatsim will wake and sleep based on the current flight context.

During departure, XVatsim may show departure airport services or CTAF/UNICOM fallback. During cruise, it usually shows only route-relevant center controllers. Near destination, it wakes for arrival when the aircraft reaches the arrival preparation window.

Understanding The Overlay

The overlay shows the flight-aware frequency board, COM radio state, transmit and receive state, Mode C state, and Standby Assist state.

Color and labels matter:

- Green row: current-polygon or current-flight authority.
- Orange row: next route polygon or arrival-prep authority.
- `Active`: the frequency is tuned in an active COM radio.
- `Standby`: Standby Assist loaded the frequency into COM1 standby.
- `ASST ON`: Standby Assist is enabled.
- `ASST OFF`: Standby Assist is disabled.
- `MODE C *Active*`: transponder Mode C is active.
- `TX` and `RX`: transmit and receive state for COM1 and COM2.
- Top-right version text: green means the installed version is current, gray

means update status is not known yet, and amber rotating version/UPDATE text means a newer XVatsim package is available.

Tuning a frequency does not make a row green by itself. XVatsim colors controller rows from route and authority context, not from radio tuning alone.

In Version 2.2.0, the Brain owns the final radio-board order, controller relevance decisions, information ORBs and drawers, update notice state, and Standby Assist target. COM1 active frequency is the only radio state that advances the next Standby Assist target; COM2 can be displayed, but it does not mark a controller row active or move the assist pointer.

When an update is available, XVatsim shows a dismissible update notice panel with the installed version, latest version, and these instructions: `Download: X-Plane.org or GitHub Releases` and

```
        Use the new Manager included in the package.
```

Manual update checks use the same notice panel for available, current, and failed results, so update status is not clipped into the bottom route/status line.



Center frequency display

METAR, ATIS, And PDC Information

The three round ORBs below the main card are independent information sources. Select an ORB to open its drawer. Select the same ORB again to close it, or select another ORB to switch drawers. Use the mouse wheel over an open drawer to read longer entries.

METAR

The METAR ORB automatically follows the primary airport chosen for the current flight context. The drawer shows the latest available VATSIM METAR and keeps a bounded recent history. To look up another airport without changing the automatic primary airport, choose `Plugins > XVatsim > Select METAR Airport...`, enter one four-letter ICAO code, and press Enter.

VATSIM ATIS

The ATIS ORB follows the primary departure or arrival airport and displays the current VATSIM voice-ATIS text when available. `NEW A`, for example, means a new Information A is unread; `INFO A` means that exact revision has been displayed. If VATSIM republishes identical Information A content, XVatsim keeps it read instead of treating it as a new message.

For another airport, choose `Plugins > XVatsim > Select ATIS Airport...`, enter one four-letter ICAO code, and press Enter. A manual lookup does not replace the automatic primary airport.

PDC And Private Messages

After a stable xPilot connection and complete flight identity are available, XVatsim receives incoming message observations through the supported path for the installed xPilot generation. xPilot 3 uses the proven low-frequency network log monitor. xPilot 4 uses the installed read-only SDK companion and can recover retained PDC/ACARS information already present when XVatsim starts. Both ordinary controller private messages and PDC/ACARS messages appear newest first in the PDC drawer. Public radio chat, broadcasts, server traffic, outgoing messages, and messages for another callsign are excluded by the Brain.

The PDC ORB shows amber `NEW` while any retained message is unread. Open the drawer to read the messages, then close it or switch to another drawer. The ORB returns to cyan `IDLE` and waits for the next message. xPilot remains the authoritative VATSIM client and should still be checked when message context or delivery is uncertain.

CTAF And UNICOM

When no controlled local airport frequency is relevant, XVatsim can display CTAF or UNICOM fallback. For airports where no CTAF is published, XVatsim can show a no-CTAF / UNICOM fallback state.

For example, in some regions a destination or departure may show `NO CTAF / UNICOM 122.800`.



European UNICOM fallback

Plugin Menu

Open X-Plane's menu bar and choose `Plugins > XVatsim`.

IFR Mode / VFR Mode

Selects the saved operating-mode preference. Version 2.0.2 retains the VFR mode foundation, but controller projection remains designed and proven for the IFR flight-plan workflow.

Manual CTAF Lookup

Opens a text prompt for looking up CTAF information manually. Enter the airport ICAO and press Enter. The prompt accepts CTAF lookup text such as `.ctaf KJAC` or simply the airport ICAO when the prompt is already open.

Select METAR Airport

Opens a text prompt for a one-airport METAR lookup. Enter one four-letter ICAO code and press Enter. The result opens in the METAR drawer without changing the automatic primary airport.

Select ATIS Airport

Opens a text prompt for a one-airport VATSIM ATIS lookup. Enter one four-letter ICAO code and press Enter. The result opens in the ATIS drawer without changing the automatic primary airport.

Open Display

Forces the overlay open. Use this when you want to see XVatsim even if Auto Display would currently keep it asleep.

Close Display

Forces the overlay closed. Use this when you want the overlay hidden.

Auto Display

Returns XVatsim to automatic wake and sleep behavior. This is the normal mode for flying.

More Opacity / Less Opacity

Adjusts overlay opacity.

Larger UI / Smaller UI

Adjusts overlay scale. XVatsim also remembers overlay size changes made by resizing the overlay window.

Faster Animation / Slower Animation

Adjusts the roll-down and roll-up animation speed.

Reset Appearance

Resets overlay opacity, scale, and animation speed to defaults.

Set Cruise Target To Current Altitude

Sets XVatsim's cruise target to the aircraft's current altitude. This can help when your flown cruise altitude differs from the filed altitude.

Reset Cruise Target To Filed Altitude

Returns the cruise target to the filed VATSIM flight-plan altitude.

Reset XVatsim Session

Clears flight-scoped XVatsim state. Use this when starting a new flight or when you intentionally want XVatsim to forget the current session.

Recover Current Flight

Recovers XVatsim workflow state for the currently active VATSIM flight plan. Use this after an xPilot disconnect/reconnect or if XVatsim needs to resume monitoring an active flight.

This is not the same as Reset XVatsim Session. Recover Current Flight is intended to keep working with the current flight plan.

Check for Updates

Runs a manual notify-only update check against the public XVatsim update manifest. If the installed version is current, XVatsim shows a short current-version status. If a newer version is available, XVatsim tells you to download it from the X-Plane.org file page or the official GitHub Releases page.

The in-simulator plugin does not download, install, replace, or launch a browser for updates. The standalone Version 2.2.0 Manager can retrieve later compatible payloads only after completing its full verification checks, and it waits for the pilot to select the offered action.

Set Diversion Airport

Opens a text prompt for setting a diversion airport. Enter a four-letter airport ICAO. XVatsim retargets arrival logic to the diversion airport when the entry is accepted.

Revert To VATSIM Flight Plan

Clears the manual diversion override and returns XVatsim to the destination in the active VATSIM flight plan.

Standby Assist On / Off

Enables or disables Standby Assist. When enabled, XVatsim can preload COM1 standby with the next frequency selected by the Brain. The order depends on the flight stage and the destination airport's resolved radio information:

- Arrival with CTAF: Center, Approach/Departure, Tower, CTAF, Ground.
- Arrival with no CTAF: Center, Approach/Departure, UNICOM `122.800`, Tower,

Ground.

- Enroute: eligible route controllers, then UNICOM `122.800` only when no

eligible Enroute controller is online.

For the Enroute fallback, COM2 already being active on `122.800` does not prevent Standby Assist from loading `122.800` into COM1 standby. COM1 active on `122.800` still prevents a redundant standby load.

For Arrival and Enroute, the normal Standby Assist switch authorizes this sequence. The older separate direct-CTAF compatibility setting continues to apply to Departure only. XVatsim does not guess UNICOM while an Arrival CTAF lookup is pending or after it fails. All targets use the same Brain-approved COM1 standby writer and existing radio safety checks. The overlay shows `ASST ON` or `ASST OFF`.

Keyboard Commands In X-Plane

XVatsim registers several X-Plane commands. You can bind these to keyboard keys, joystick buttons, or hardware controls.

To assign a keyboard command:

1. Open X-Plane.
2. Go to `Settings`.
3. Open the `Keyboard` tab.
4. Search for `xvatsim`.
5. Select the command you want.
6. Assign a key or button.
7. Click `Apply`.

Bindable XVatsim commands:

X-Plane command	Purpose
`xvatsim/manual_ctaf_lookup`	Open the manual CTAF lookup prompt.
`xvatsim/display_open`	Force the XVatsim display open.
`xvatsim/display_close`	Force the XVatsim display closed.
`xvatsim/display_auto`	Return the display to automatic behavior.

X-Plane command	Purpose
`xvatsim/cruise_target_current`	Set the cruise target to current aircraft altitude.
`xvatsim/cruise_target_filed`	Reset the cruise target to filed VATSIM altitude.
`xvatsim/reset_session`	Reset XVatsim state for the next flight.
`xvatsim/recover_current_flight`	Recover XVatsim workflow state for the current flight.

Menu-only functions, such as appearance controls, diversion airport entry, and Standby Assist On/Off, are controlled from `Plugins > XVatsim`.

Long-Haul And Reconnect Use

If xPilot disconnects during a long flight, reconnect to VATSIM with the same callsign and active flight plan. Then use:

```
Plugins > XVatsim > Recover Current Flight
```

XVatsim will attempt to reload the active VATSIM flight plan and resume monitoring the current flight. If the plan is missing, stale, or does not match the current callsign, recovery may fail closed rather than guessing.

When To Use Reset Session

Use `Reset XVatsim Session` when:

- You are starting a new flight.
- You changed aircraft or callsign and want a clean XVatsim state.
- You intentionally want XVatsim to forget the current flight.

Do not use Reset Session just to recover after a reconnect. Use `Recover Current Flight` for that.

Troubleshooting

XVatsim menu does not appear

- Confirm the plugin file exists at `X-Plane 12\Resources\plugins\XVatsim\win_x64\XVatsim.xpl`.
- Confirm you are running X-Plane 12 on Windows.
- Check `X-Plane 12\Log.txt` for plugin load errors.

Overlay does not appear

- Start xPilot and connect to VATSIM.
- Confirm the aircraft has valid electrical/radio state.
- Choose `Plugins > XVatsim > Open Display`.
- Choose `Plugins > XVatsim > Auto Display` to return to normal behavior.

Flight plan does not load

- Confirm the VATSIM callsign in xPilot matches the filed flight plan.
- Give XVatsim time to refresh after connecting.
- Use `Plugins > XVatsim > Recover Current Flight`.

Wrong or unexpected frequency display

- Compare what xPilot shows with what XVatsim shows.

- Remember that XVatsim intentionally hides irrelevant or unproven controllers.
- Current controllers display green; next or arrival-prep controllers display orange.
- If a relevant controller is missing or an irrelevant controller appears, collect logs and report it.

Standby Assist did not tune a frequency

- Confirm Standby Assist is on.
- For controller targets, confirm a relevant live controller exists.
- For CTAF/UNICOM advisory targets, wait for the lookup to finish.
- Confirm the recommended frequency is not already active.
- On Arrival, check the applicable sequence above. At a no-CTAF airport,

UNICOM intentionally comes before Tower and Ground.

- Enroute UNICOM is used only when no eligible Enroute controller is online.
- The PDC information drawer is separate from Standby Assist and never tunes a frequency.

METAR, ATIS, or PDC information is missing

- Confirm xPilot is connected and the flight callsign matches the filed plan.
- METAR and ATIS availability depends on the current VATSIM sources.
- For a different airport, use `Select METAR Airport...` or `Select ATIS Airport...` from the XVatsim plugin menu.
- Confirm xPilot is connected with the active callsign. For xPilot 4, run the

Manager and choose Repair if the companion is missing or damaged.

- Unchanged ATIS content stays read when VATSIM republishes it.

Update check is unavailable

- Confirm you have internet access.
- Confirm the public update file is reachable at `https://ezsimulations.github.io/XVatsim/xvatsim_update.json`; a temporary failure affects only update checks.

Bug Reports

For useful bug reports, include:

- Departure airport.
- Arrival airport.
- Callsign.
- Filed route if available.
- What xPilot showed.
- What XVatsim showed.
- Screenshots if possible.
- Whether Standby Assist was on or off.
- Whether you had disconnected/reconnected xPilot.
- The files listed below.

Fresh diagnostic logs are generated locally when the plugin runs. These logs are not shipped inside the package, but they are useful for troubleshooting.

Log files:

```
X-Plane 12\Resources\plugins\XVatsim\logs\xvatsim_diagnostics_YYYY_MM_DD.log
```

```
X-Plane 12\Log.txt
```

Support contact:

```
ezsimulations@gmail.com
```

Freeware Notes

XVatsim is being provided as freeware. Please keep the package intact when sharing it so pilots receive the Manager, plugin, transition audio, authority registry, README, quick start, checksum file, and this user guide together. XVatsim 2.2.0 is focused on Windows, X-Plane 12, xPilot, and IFR flight-plan operations.